

ECE448 Spring 2026

L04-sdr-

introduction

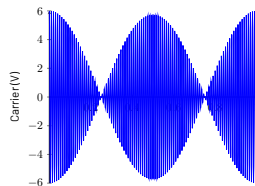
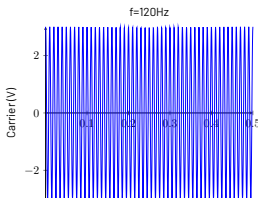
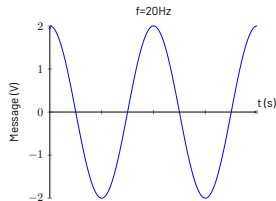
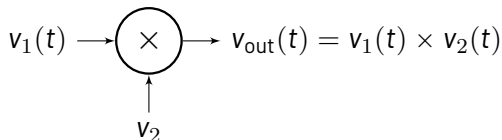


UNITED STATES
AIR FORCE
ACADEMY

TODAY

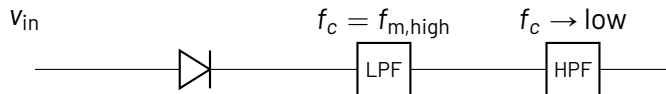
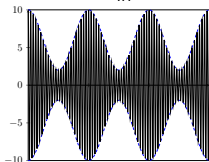
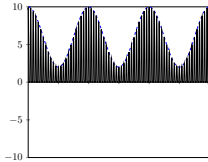
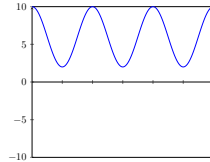
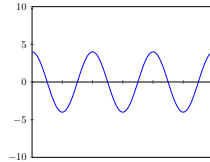
- Software Defined Radios
- Superheterodyne Architecture
- IQ data
- Examples

FUNCTION MULTIPLIER

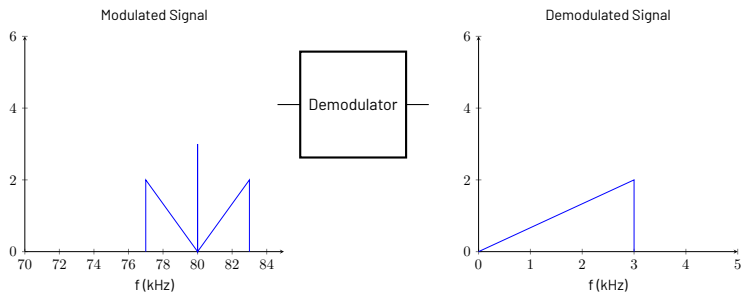


$$v_m(t) \times v_c(t) = v_{\text{out}}$$

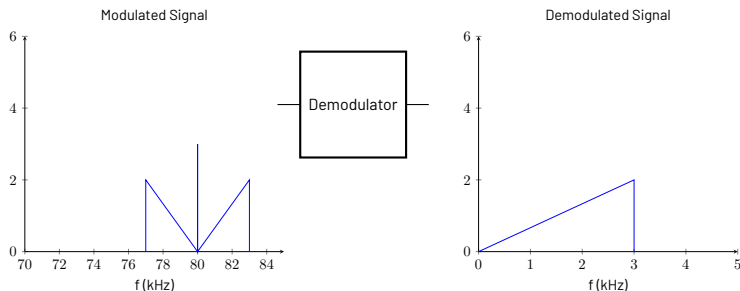
ENVELOPE DETECTOR

 V_{in}  $V_{out, \text{diode}}$  $V_{out, \text{LPF}}$  $V_{out, \text{HPF}}$ 

SYNCHRONOUS DETECTOR

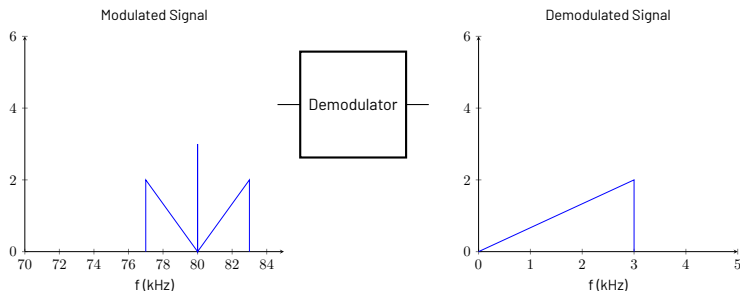


SYNCHRONOUS DETECTOR



$$\alpha = \frac{A_m}{B} = \frac{2 * \left(\frac{A_c A_m}{2} \right)}{B A_c} = \frac{2 * 2}{3} = 1.333$$

SYNCHRONOUS DETECTOR



$$\alpha = \frac{A_m}{B} = \frac{2 * \left(\frac{A_c A_m}{2} \right)}{B A_c} = \frac{2 * 2}{3} = 1.333$$

The signal is overmodulated → need synchronous detection!

WHAT IS AN SDR?

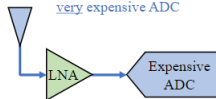
- “A radio in which *some* or *all* of the physical layer is defined by software”
- Characteristics:
- Applications



RECEIVER ARCHITECTURES: DIRECT SAMPLING/ZEROIF

Direct Sampling (a.k.a. Direct RF)

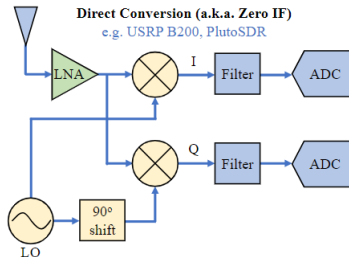
very expensive ADC



<https://wirelesspi.com/direct-conversion-zero-if-receiver/>

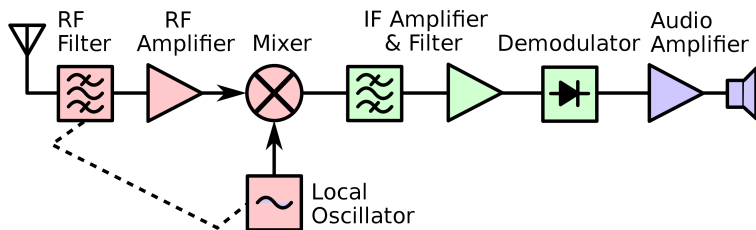
Direct Conversion (a.k.a. Zero IF)

e.g. USRP B200, PlutoSDR



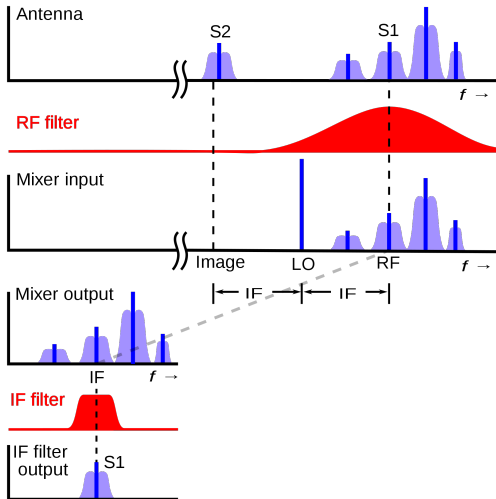
<https://wirelesspi.com/direct-conversion-zero-if-receiver/>

SUPERHETERODYNE RECEIVER



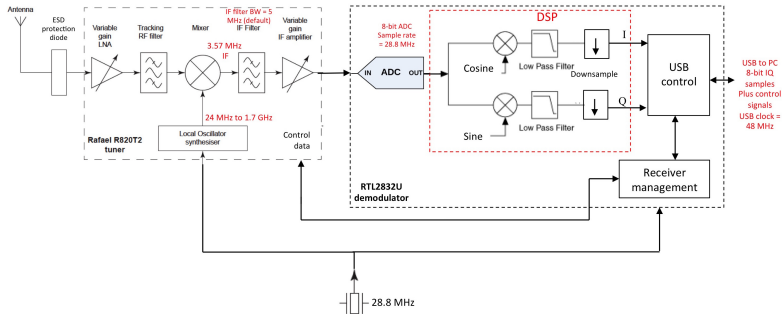
<https://wirelesspi.com/the-heterodyne-principle-and-the-superheterodyne-receiver/>

SUPERHETERODYNE RECEIVER



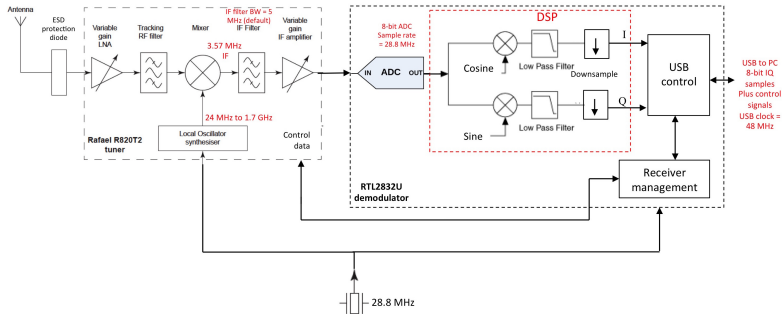
RTL-SDR RECEIVER

Simplified Block Diagram of NooElec RTL-SDR



RTL-SDR RECEIVER

Simplified Block Diagram of NooElec RTL-SDR



RTL-SDR



RTL-SDR

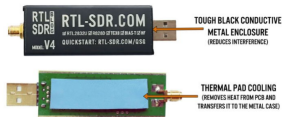
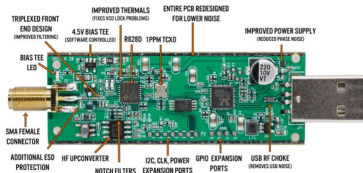
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CHOOSE A GENUINE RTL-SDR BLOG V4



ETTUS X310



SDRs: PRACTICAL QUESTIONS

- Frequency shift
- Negative Frequencies?
- Waterfall plot
- Tradeoffs
- Software

SOFTWARE

- GNURadio

- GQRX

- SDR Angel

- SDR #

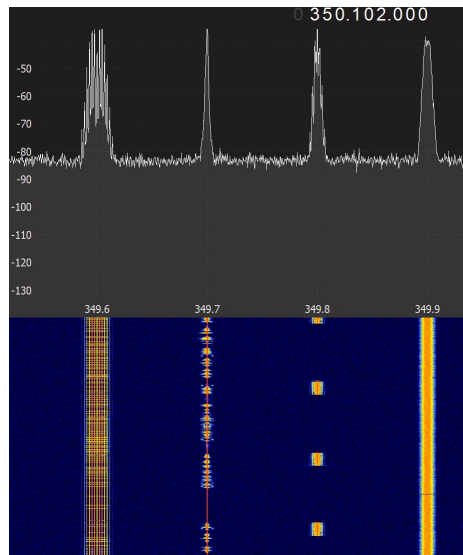
- Sigdigger

- URH



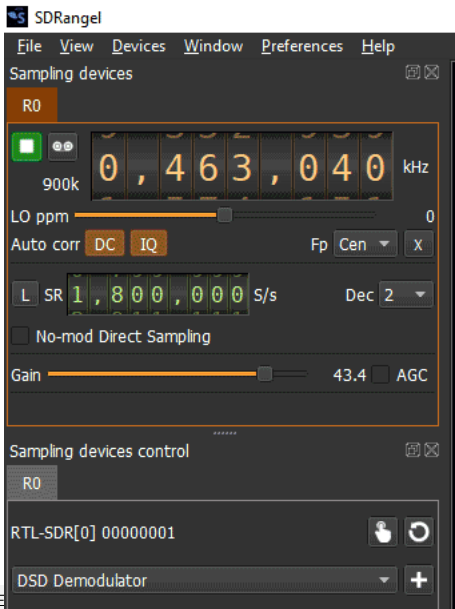
SOFTWARE

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SOFTWARE

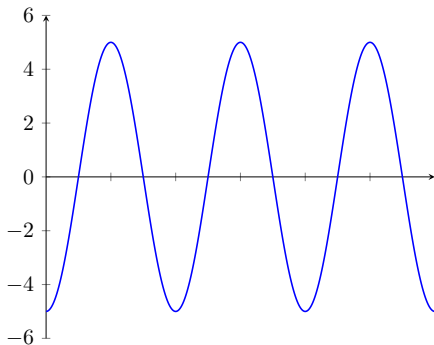
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TODAY

- Software Defined Radios
- IQ data
- Examples

IQ DATA

$$x(t) = \cos(2\pi ft + \phi)$$



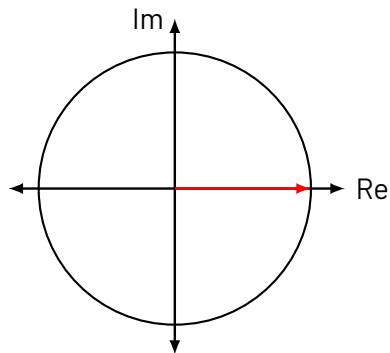
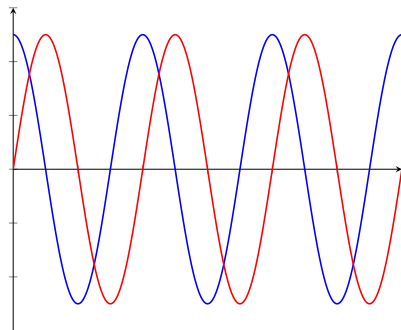
IQ data is a way to transmit both _____ and _____ without _____.

$$\left(\sqrt{I^2 + Q^2}\right) \cos \left[2\pi ft + \tan^{-1} \left(\frac{Q}{I}\right)\right]$$

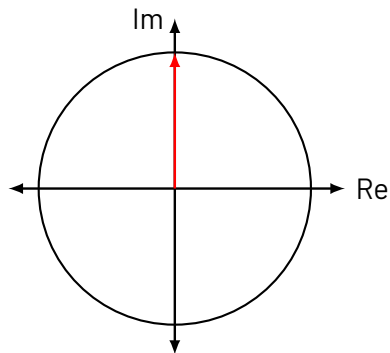
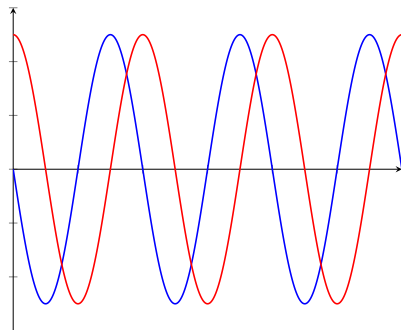
COMPLEX PLANE

IQ DEMO

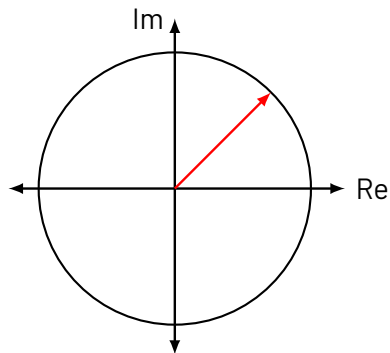
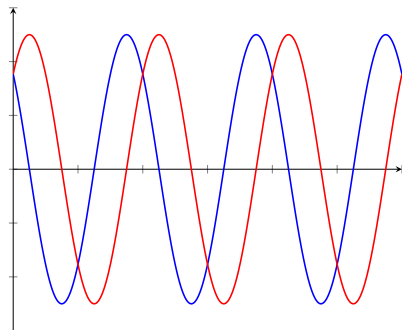
COMPLEX PLANE

 $I =$  $Q =$

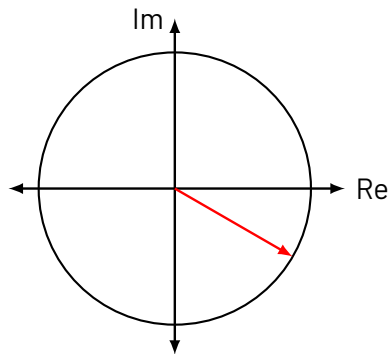
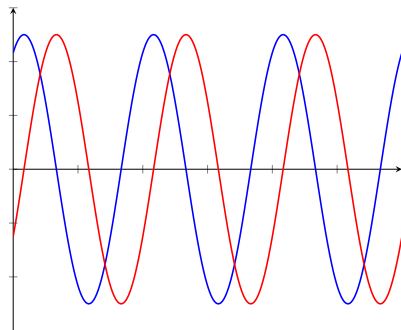
COMPLEX PLANE

 $I =$  $Q =$

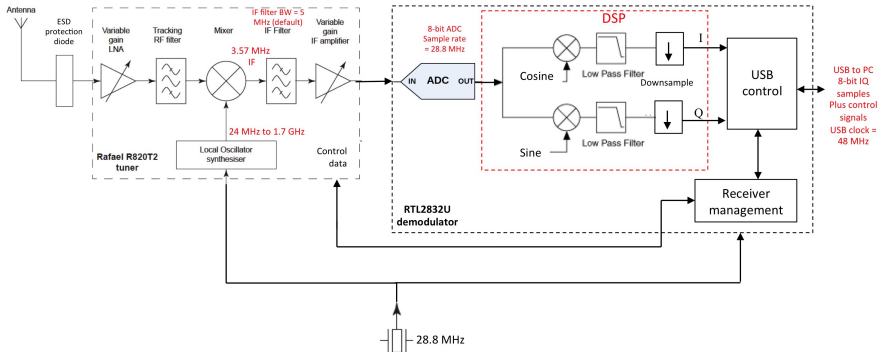
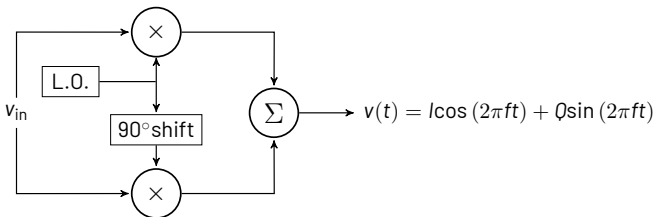
COMPLEX PLANE

 $I =$  $Q =$

COMPLEX PLANE

 $I =$  $Q =$

INSIDE THE SDR



WORKING WITH IQ DATA

- IQ data is stored as pairs of 32-bit floats
- Working with data in Python:

```
import numpy as np
import matplotlib.pyplot as plt
f = np.fromfile(open( 'filename', 'r' ), dtype=np.float32)
plt.plot(f)
plt.show()
```

- Average power of a digital signal is:

$$P = \frac{1}{N} \sum_{n=1}^N |x[n]|^2$$

- How would we calculate PSD in Python?

POWER SPECTRAL DENSITY

- Take the FFT of a selection of x samples. The output will be x -long array of complex floats.
- Take the magnitude of the FFT output, which provides us 1024 real floats.
- Square the resulting magnitude to get power.
- Normalize: divide by the FFT size (x) and sample rate (F_s).
- Convert to dB using $10 \log_{10}()$
- Perform an FFT shift

SDRs AND SAMPLING

- You already know about Nyquist
- But what does this mean for SDRs?
- Let's investigate w/ GNURadio...

DEMO

GNURADIO

The following instructions come from [GNURadio Wiki](#):

- Install miniforge (which is a conda install)
- Open the conda prompt
- Install mamba: `conda install mamba`
- Create a conda env: `mamba create -n gnuradio`
- Activate the env: `mamba activate gnuradio`
- Configure conda channels
`mamba config --env --add channels conda-forge`
`mamba config --env --set channel_priority strict`
- Install gnuradio `mamba install gnuradio`
- Upgrade all your packages: `mamba upgrade --all`
- You'll want UHD: `mamba install gnuradio-uhd`
- There are LOTS of additional packages

TODAY

- GNURadio Overview
- Sinks/Sources
- Data types
- Hardware
- Simple receiver example

GETTING STARTED

- In your Windows menu, search for mamba prompt
- See the environments installed: mamba env list
- Activate radioconda: mamba activate gnuradio
- Start gnuradio: gnuradio-companion

```
Conda Prompt - gnuradio-companion
(base) C:\Users\neil.rogers>conda env list
# conda environments:
#
base                  *  C:\Users\neil.rogers\Anaconda3
                       *  C:\Users\neil.rogers\AppData\Local\radioconda

(base) C:\Users\neil.rogers>mamba env list
# conda environments:
#
base                  *  C:\Users\neil.rogers\Anaconda3
                       *  C:\Users\neil.rogers\AppData\Local\radioconda

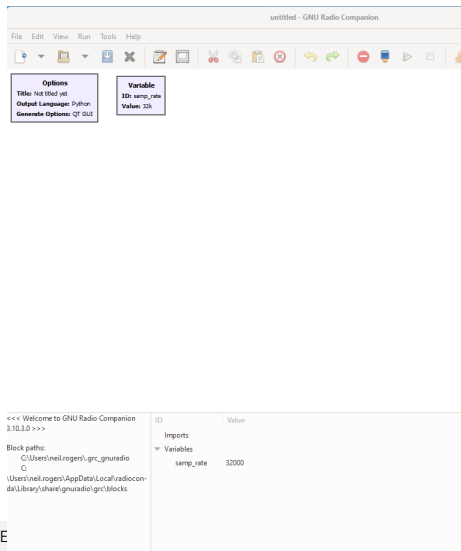
(base) C:\Users\neil.rogers>mamba activate base

(base) C:\Users\neil.rogers>gnuradio-companion
>>> Warning: vocoder_codec2_decode_ps - option_attributes are for enums only, ignored
>>> Warning: vocoder_codec2_encode_sp - option_attributes are for enums only, ignored
<<< Welcome to GNU Radio Companion 3.10.3.0 >>>

Block paths:
  C:\Users\neil.rogers\grc_gnuradio
  C:\Users\neil.rogers\AppData\Local\radioconda\Library\share\gnuradio\grc\
```

GETTING STARTED

- In your Windows menu, search for mamba prompt
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`mamba activate gnuradio`
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`gnuradio-companion`



MAIN WINDOW

untitled - GNU Radio Companion

File Edit View Run Tools Help

Options
Title: Not titled yet
Output Language: Python
Generate Options: QT GUI

Variable
ID: samp_rate
Value: 32k

Core

- Audio
- Boolean Operators
- Byte Operators
- Channel Models
- Channelizers
- Coding
- Control Port
- Debug Tools
- Deprecated
- Digital Television
- Equalizers
- Error Coding
- File Operators
- Filters
- Fourier Analysis
- GUI Widgets
- Impairment Models
- Industrial I/O
- Instrumentation
- IQ Correction
- Level Controllers
- Math Operators
- Measurement Tools
- Message Tools
- Misc
- Modulators
- Networking Tools
- OFDM
- Packet Operators
- PDU Tools
- Peak Detectors

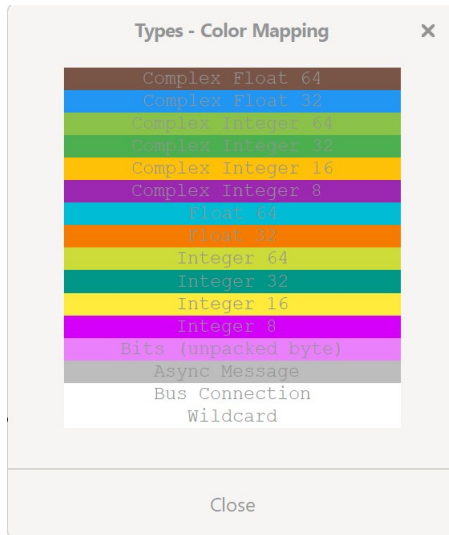
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Block paths:

- C:\Users\neil.rogers\gnuradio
- C:\Users\neil.rogers\AppData\Local\radiocompanion\Library\share\gnuradio\grc\blocks

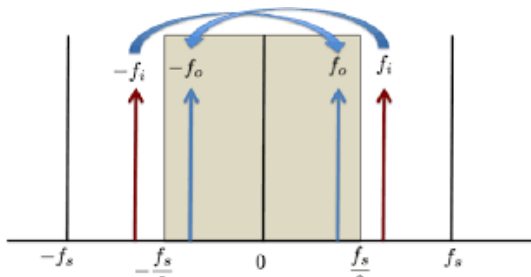
ID	Value
Imports	
Variables	
samp_rate	32000

DATA TYPES



SAMPLING

- We all know Nyquist, but what happens when we violate it?
- Remember, we are dealing with digital signals
- Analog Devices



HARDWARE

- Hackrf → check version with `hackrf_info`
- UHD → `uhd_usrp_probe`
- GQRX
- NBFM